7.1 Normal vector and Cartesian equation

Definition and results

A nonzero normal vector to a line is orthogonal to every direction vector of that line. If $\vec{n} = (a;b) \neq (0;0)$ is normal and $A(x_A;y_A)$ belongs to d, then d has equation $a(x-x_A)+b(y-y_A)=0$, that is $ax+by+c=0$. A possible direction vector is $(-b;a)$.

Status and justification. $M \in d$ is equivalent to $\overrightarrow{AM}\cdot\vec{n}=0$. In an orthonormal frame, expanding this product gives the equation. Conversely, the set so defined is the line through A orthogonal to n.

Worked example. Through $A(1;2)$ with normal $(2;-3)$: $2(x-1)-3(y-2)=0$, so $2x-3y+4=0$. The vector $(3;2)$ directs the line: product $6-6=0$.

Watch out

$(a;b)$ is normal, not a direction vector in general. $(a;b)$ must not be zero.

Immediate application. Equation of the line through $(2;-1)$ with normal $(3;4)$.

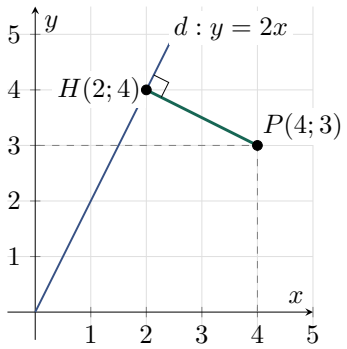
7.2 Orthogonal projection and distance

Definition and results

The projection H of P onto d is characterised by $H \in d$ and $\overrightarrow{PH}$ collinear with a normal vector of d. For $d: ax+by+c=0$, write $H = P-\lambda(a;b)$, with $\lambda = \dfrac{ax_P+by_P+c}{a^2+b^2}$. The distance is $PH = \dfrac{|ax_P+by_P+c|}{\sqrt{a^2+b^2}}$. The distance formula is a deduced supplement, not an isolated routine.

Status and justification. Substitute $x_H = x_P-\lambda a$, $y_H = y_P-\lambda b$ in the equation: $ax_P+by_P+c-\lambda(a^2+b^2)=0$. The denominator is positive; we obtain H, then $PH = |\lambda|\sqrt{a^2+b^2}$.

Worked example. For $P(3;4)$ and $d: x+y-2=0$, $\lambda = \dfrac{3+4-2}{2} = \dfrac{5}{2}$, so $H(\frac{1}{2};\frac{3}{2})$ and $PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H+y_H=2$.



The segment PH is perpendicular to d. Both axes have the same scale; the coordinates of P and H can be read directly.

Watch out

The distance is positive or zero: keep the absolute value. Two checks, membership and perpendicularity, secure the projection.

Immediate application. Project $P(4;1)$ onto $d : x = 2$.

7.3 Equation of a circle

Definition and results

The circle with centre $I(\alpha;\beta)$ and radius $r > 0$ has equation $(x-\alpha)^2 + (y-\beta)^2 = r^2$. An equation $x^2 + y^2 + ux + vy + w = 0$ is handled by completing both squares. If the squared radius obtained is negative, the set is empty; if it is zero, it is a point.

Status and justification. M belongs to the circle if and only if $IM = r$. Since both sides are positive or zero, we can square and use the distance formula in an orthonormal frame.

Worked example. $x^2 + y^2 - 4x + 6y + 9 = 0$ becomes $(x-2)^2 + (y+3)^2 = 4$. Centre $(2;-3)$, radius 2. The signs of the coordinates of the centre are opposite to those appearing in the brackets.

Watch out

A quadratic equation in x and y is not always a circle: check the coefficients and the real radius.

Immediate application. Recognise $x^2 + y^2 + 2x - 8y + 17 = 0$.

7.4 Solving a configuration

Definition and results

An intersection is obtained by solving the equations simultaneously. For a circle and a line parallel to an axis, substituting the constant coordinate gives a quadratic equation. For a perpendicular bisector, write the equality of the squared distances; the quadratic terms cancel.

Status and justification. The equivalences rest on membership of both sets. The equality $MA = MB$ is equivalent to $MA^2 = MB^2$, since the distances are positive.

Worked example. The circle $x^2 + y^2 = 25$ and the line $x = 3$ meet at $(3; 4)$ and $(3; -4)$. The perpendicular bisector of $A(0; 0)$ and $B(4; 2)$ satisfies $x^2 + y^2 = (x - 4)^2 + (y - 2)^2$, so $2x + y = 5$.

Watch out

An intersection is not the union of the solution sets. Check the points in both equations.

Immediate application. Intersect $(x - 1)^2 + (y + 2)^2 = 9$ with $y = 1$.